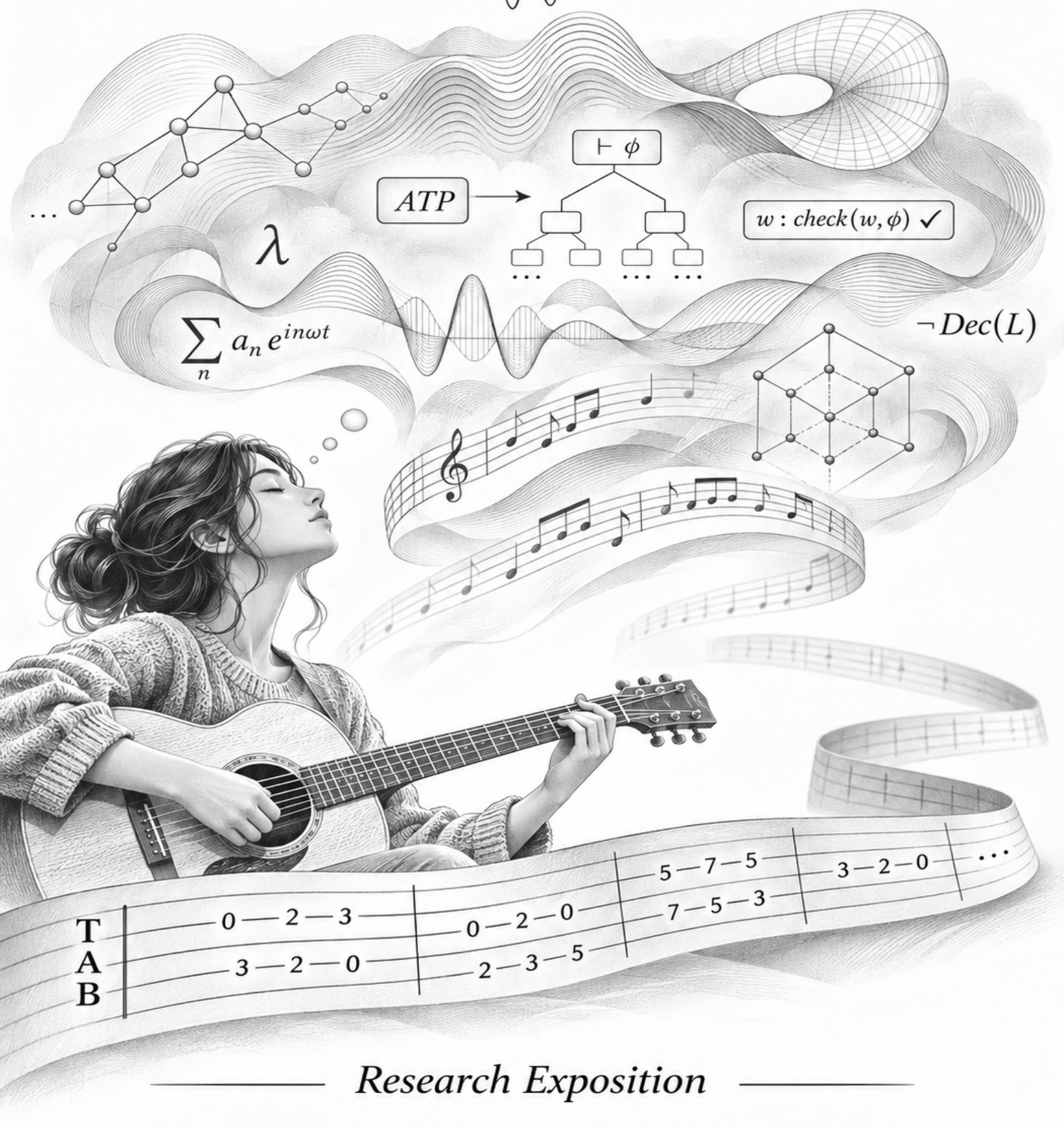
Undecidable Attained Labels with Finite Checkable Witnesses



Research Exposition

Undecidable Attained Labels with Finite Checkable Witnesses

Five-Party Quantum Separation, Deterministic Realization Limits, and Verifier-Gated Adaptive Composition

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Research conception; human-directed, AI-assisted mathematical exposition

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Abstract

We integrate the mathematical and research questions recorded on 3 November 2026 concerning possible ATP dreams (MUSIC), legally coordinated proof performances (SYMPHONY), deterministic–stochastic interfaces (DNI), closure-relative Multiarticle transcendence, and the proposed composer–conductor role of Player Zero. We carefully separate prior constructive results from new finite quantum calculations, standard background theorems, and open research goals. The central five-qubit example has identical proper reduced states and full support under local Pauli-X measurements but different joint probability distributions at total variation distance $1/2$; consequently, unweighted MUSIC and unweighted score support cannot determine the law. A finite deterministic realization with randomness only at initialization cannot generate all finite histories of a repeated full-support 32-symbol process. We also give the precise strong-lumpability criterion, distinguish effective composition from incomputable random outputs and undecidable *attainability*, and prove a conditional conservativity invariant for adaptively composed proof searches. Crucial limits remain: no complete-source Multiarticle transcendence certificate, independently installed five-ATP federation, general complexity separation, or new physical prediction is established. A prominent research program states the evidence and verification obligations for pursuing these possibilities.

Keywords. automated theorem proving; cooperative dreaming; MUSIC; SYMPHONY; DNI; quantum parity; strong lumpability; Multiarticle theory; provenance; adaptive composer; Player Zero.

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1 Purpose, provenance, and the central question

The research collection connects five steps: (i) individual possible dream histories; (ii) authorized performances of five cooperating ATPs; (iii) compact, certified proof-search representations; (iv) deterministic and stochastic interfaces, including a quantum toy model; and (v) the possibility that Player Zero revises future search plans as well as conducting current ones. The source collection cites the author's *MUSIC of an Automated Theorem Prover* [1], *Symphonies of Automated Theorem Provers* [2], a deterministic–nondeterministic interfaces reader [3], and *Multiarticle Theory: A Short Foundational Text* [4]. The earlier *From MUSIC to Emergent Symphonic Compression* working exposition is the immediate source of the five results recalled in Section 3 [5].

Provenance contract. In this revised edition, a *definition* stipulates an object, a *mathematical theorem* follows from explicitly stated premises, a *source-derived result* has antecedent exposition elsewhere, an *empirical proposition* needs measured hardware or a deployed implementation, and a *research conjecture* remains unsettled. The five role labels P, R, D, I, C describe an AMLD+0-style analytical protocol in the source discussion. They are not evidence of five independent theorem-proving agents having executed these proofs.

The desired physics application is a narrowly formalized interface between quantum measurement statistics and relativistic causal constraints. The phrase *Einstein–Peres interface* is a research label here, not an assertion that general relativity has been reconciled with quantum theory or that an equation from either historical author is proved by this paper.

2 Frozen mathematical objects and their types

2.1 MUSIC: individual possibility without a canonical law

Fix a complete ATP condition c, including theory, certified BANK, internal players, legal game rules, permitted reports, and an initial or intermediate state. Let $\mathcal{H}_c$ be the set of its legal episodes, and let

the typed location reports in episode h form $R(h)$. Set

$$d(h) = \bigcup_{r \in R(h)} r, \qquad \mathcal{D}_c = \{d(h) : h \in \mathcal{H}_c\}, \qquad \mathrm{MUSIC}(c) = \mathcal{P}(\mathcal{D}_c). \tag{1}$$

As stipulated in the author's earlier definition [1], the right-hand object is the *set of all sets* of individually possible report-union dreams. It contains no canonical probability measure. In particular, $\{d_1, d_2\} \in \mathrm{MUSIC}(c)$ does not imply that a single legal run can produce both dreams. Untagged unions can also forget order, repetition, and provenance; relevant historical predicates must be preserved in reports or authenticated sidecars.

2.2 SYMPHONY: lawful coordination and evidence-bearing scores

Let F be a five-ATP federation and $\mathrm{Runs}(F, c)$ the set of legal runs from its frozen checkpoint. A structured score $S(\rho)$ may retain the five local projections, event/causal order, authenticated transfers, coordinator actions, and independent checker decisions. Define

$$\mathrm{SYMPHONY}(F, c) = \{S(\rho) : \rho \in \mathrm{Runs}(F, c)\}. \tag{2}$$

A score can be evidence-preserving without being an inference rule. An *unweighted* set of scores is not a probability distribution: a stochastic version requires a specified experiment $\Pr_c$ on legal runs, a measurable score space, and the pushforward measure $\mu_c = S_* \Pr_c$. A complete information contract must also state which events the encoding can decode, and which verifier artifacts remain separately accessible [2].

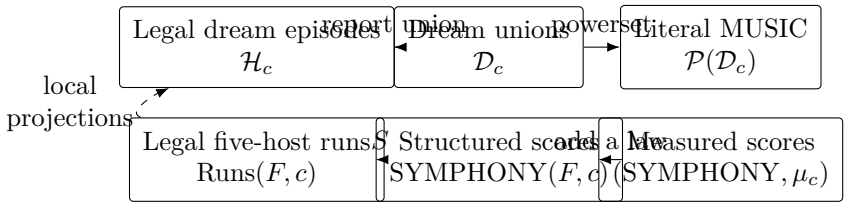


Figure 1: Three distinct kinds of information: individual dream support, ordered legal coordination, and a probability law on scores. Dashed projection is not an equality of mathematical types.

2.3 Multiarticle transcendence: relative closure, not search failure

The working research record attributes a closure-relative comparison to *Multiarticle Theory Foundations 0.1* [4]. For source articles $A_i = (L_i, F_i, \Gamma_i, G_i, V_i, \nu_i)$ and declared meaning-preserving translations τ_i into a fixed comparison calculus F, write

$$C_i = \mathrm{Cn}_F(\tau_i(\Gamma_i)), \qquad C_M = \mathrm{Cn}_F\left(\bigcup_i \tau_i(\Gamma_i) \cup B\right), \qquad \mathrm{Tr}(M; A_1, \ldots, A_k) = C_M \setminus \bigcup_i C_i. \tag{3}$$

The formula is reproduced from the attached collection's description of Foundations 0.1; the complete original formal source and all translations have *not* been audited in this revision. A target T meets this displayed criterion only when $T \in C_M$ and $T \notin C_i$ for *every* i. Failing to find an individual proof is not proof of individual non-derivability. In classical explosive calculi, inconsistent joint premises also require special scrutiny.

Remark 2.1 (Bridge-only novelty is not genuine cross-source synergy). Even if $T \in C_M \setminus \bigcup C_i$, one could obtain this vacuously by taking $B = \{T\}$ and letting all source theories be empty. A useful stronger synthesis claim therefore demands an independently justified bridge and source-contribution tests; the displayed set-difference criterion alone does not enforce those methodological obligations.

3 Five precursor constructions and their exact limits

The source collection recalls five results from the author's earlier emergent-compression exposition [5]; these are *not* new theorem-priority claims in the present paper.

E1. Five-essential Horn evidence with cheap operational quotient. For each $n \geq 1$, each of five hosts holds $E_i = \{a_{i,k} : 1 \leq k \leq n\}$. The shared Horn rules are

$$\Gamma_n = \left\{ \left(\bigwedge_{i=1}^{5} a_{i,k} \right) \to b_k : 1 \leq k \leq n \right\} \cup \left\{ \left(\bigwedge_{k=1}^{n} b_k \right) \to g \right\}. \tag{4}$$

No proper coalition entails g from its authorized premise sets, while all five collectively derive g in $O(n)$ proof steps under the stated representation. Each host can make 2^n distinct binary-report dreams, hence has 2^{2^n} MUSIC members. The prescribed nonmutating protocol can nevertheless have an $O(n)$-state quotient. This proves *evidence essentiality*, not an unconditional five-processor complexity advantage.

E2. Local non-reconstruction. Two federations can possess the same local pre-transfer MUSIC objects, yet differ in global reachability of g, because only one permits authorized evidence pooling. The comparison fixes local conditions *before* transfer and changes the global authorization interface.

E3. Affine-history identification under a promise. If a trusted specification guarantees that the Boolean terminal map $V : \mathbb{F}_2^m \to \mathbb{F}_2$ is affine, its m coefficients and offset follow from the $m + 1$ queries $0, e_1, \ldots, e_m$:

$$V(x) = c \oplus \bigoplus_{j=1}^{m} \alpha_j x_j, \quad c = V(0), \quad \alpha_j = V(e_j) \oplus V(0). \tag{5}$$

The continuation key $(t, c \oplus \bigoplus_{j \leq t} \alpha_j x_j)$ has at most $2(m + 1)$ states, with $O(m^2)$ bits of complete experiment recording. Discovering this *terminal-behavior* quotient is distinct from reconstructing checkable proof certificates, which needs additional interfaces and scope assumptions.

E4. Unrestricted black-box obstruction. For an arbitrary Boolean function on $\{0, 1\}^m$, fewer than 2^m distinct exact queries leave an unqueried input whose label could be changed without changing the transcript. Thus universally exact reconstruction cannot be guaranteed by polynomially many black-box observations absent further structure.

E5. Centralized simulation. Five fixed effective processes of polynomial aggregate work, communication, and certificate checking can be simulated with polynomial overhead by one machine given the same inputs, random choices, representations, and authorized responses. This does not preclude practical gains from parallel hardware, restricted authority, or better heuristics; it rules out an unqualified polynomial-versus-superpolynomial separation against an unrestricted full-information simulator under those assumptions.

4 Five-party quantum witness: equal supports, different laws

Theorem 4.1 (Full-support five-party statistical separation)**.** *There exist five-qubit density matrices ρ_+ and ρ_- such that (i) their reduced states on* every proper subsystem *agree; (ii) all 32 outcomes of local Pauli-X measurements have positive probability for both preparations; and (iii) their full five-party outcome distributions have total variation distance* $1/2$.

Proof. Write $|\mathbf{0}\rangle = |00000\rangle$, $|\mathbf{1}\rangle = |11111\rangle$. For $\sigma \in \{-1, +1\}$ define

$$\rho_\sigma = \frac{1}{2}\left(|\mathbf{0}\rangle\langle\mathbf{0}| + |\mathbf{1}\rangle\langle\mathbf{1}|\right) + \frac{\sigma}{4}\left(|\mathbf{0}\rangle\langle\mathbf{1}| + |\mathbf{1}\rangle\langle\mathbf{0}|\right). \tag{6}$$

On $\mathrm{span}\{|\mathbf{0}\rangle, |\mathbf{1}\rangle\}$ this is $\left(\begin{smallmatrix} 1/2 & \sigma/4 \\ \sigma/4 & 1/2 \end{smallmatrix}\right)$, with eigenvalues $3/4$ and $1/4$. On the orthogonal complement it vanishes. Hence ρ_σ is positive semidefinite with unit trace.

Trace out at least one qubit. The off-diagonal operator $|\mathbf{0}\rangle\langle\mathbf{1}|$ vanishes under that partial trace, because one traced tensor factor contributes $\langle 1|0\rangle = 0$, as does its adjoint. Thus the proper reduced density matrices coincide.

Let $x = (x_1, \ldots, x_5) \in \{-1, +1\}^5$ describe one outcome of a product Pauli-X measurement, and put $P(x) = \prod_i x_i$. The rank-one product projector is

$$\Pi_x = \bigotimes_{i=1}^{5} \frac{I + x_i X_i}{2}, \qquad p_\sigma(x) = \mathrm{Tr}(\rho_\sigma \Pi_x) = \frac{1}{32}\left(1 + \frac{\sigma}{2} P(x)\right). \tag{7}$$

To verify the second identity, each diagonal term in (6) contributes $1/64$ after its weight, for a total $1/32$; the two off-diagonal terms together contribute $\sigma P(x)/64$. For $P(x) = \sigma$, the outcome probability is $3/64$; for $P(x) = -\sigma$ it is $1/64$. Both are positive. Any proper subset of outcomes is uniform because summing over one omitted coordinate cancels the parity factor.

There are 16 strings of each parity. Consequently,

$$\mathbb{E}_{p_\sigma} P = \frac{\sigma}{2}, \qquad \Pr_{p_+}(P = +1) = \frac{3}{4}, \quad \Pr_{p_-}(P = +1) = \frac{1}{4}, \qquad \|p_+ - p_-\|_{\mathrm{TV}} = \frac{1}{2}\sum_x |p_+(x) - p_-(x)| = \frac{1}{2}. \tag{8}$$

These identities prove all three claims. □

Table 1: Exact rational distribution for the complete five-party X measurement. There are 16 outcomes per parity sector.

Parity sector	Number of strings	Each $p_+(x)$	Each $p_-(x)$
$P(x) = +1$	16	3/64	1/64
$P(x) = -1$	16	1/64	3/64
Total probability	32	1	1

Corollary 4.2 (Support does not determine the probability law)**.** *If a fixed reporting convention records measurement outcomes without preparation tags or probabilities, the two preparations induce identical sets of possible report-union dreams and hence equal unweighted MUSIC. Their sets of legally realizable outcome-labeled scores may likewise agree. Neither common support determines both different joint distributions.*

Proof. Equation (7) is strictly positive on every outcome for both preparations, so the supported outcome sets agree. Equal supports have equal powersets. A single-valued reconstruction map on an equal input cannot return both p_+ and p_-, which differ by Theorem 4.1. □

Remark 4.3 (What the example does and does not say). The parity observable P makes the two laws distinguishable, and under equal preparation priors a single full X measurement followed by parity-based decision succeeds with probability $3/4$. The witness belongs to standard finite-dimensional quantum mathematics, not a demonstrated ATP execution or new physical prediction. The key lesson is that *the same legal possibilities can carry different measures.* A probabilistic SYMPHONY requires a specified measure, not just a richer name for the same score support.

5 The deterministic–stochastic interface: a precisely bounded obstruction

Definition 5.1 (Finite deterministic initial-seed model)**.** A deterministic hidden-state realization consists of a finite set H, an initial distribution on H, a deterministic update $f : H \to H$, and an output map $o : H \to Y$. All future observable outputs are fixed by the sampled initial hidden state; no fresh random input is allowed after initialization. Equality of processes means equality of all finite observable-word distributions, not merely some selected trajectories.

Theorem 5.2 (Finite exact realization obstruction)**.** *Let $Y = \{-1, +1\}^5$ and let $K_\sigma(y, z) = p_\sigma(z)$ for all $y, z \in Y$, where p_σ is the full-support distribution in* (7)*. No finite deterministic initial-seed model realizes this Markov chain's complete finite-word law for arbitrarily long observation horizons.*

Proof. Every symbol of Y has positive probability from every prior symbol. Thus each of the $|Y|^n = 32^n$ output words of length n has positive probability (for the stationary initial law, and also conditionally on any previous observed state). Each fixed initial hidden state of a deterministic model produces *one* length-n word, hence a model with $|H| = M$ hidden initial states supports at most M such words. Exact equality at horizon n requires $M \geq 32^n$ for every n, impossible for finite M. □

Remark 5.3 (Scope). The theorem does not prohibit a deterministic description with an *infinite* initial hidden-state space, an externally supplied random stream, nonstationary models with growing state resources, or approximate matching on a bounded horizon. It is not a theorem of physical indeterminism and does not contradict known deterministic models of quantum phenomena with different assumptions. Its contribution here is an exact, finite-model support-counting obstruction of the type discussed in the author's DNI source [3].

Theorem 5.4 (Finite strong-lumpability criterion; standard background)**.** *Let H and D be finite state spaces, $q : H \twoheadrightarrow D$ a surjection, and P a time-homogeneous Markov transition matrix on H. A Markov transition kernel $\bar{P}$ on D satisfying the* strong *projection condition for every concrete initial state exists if and only if, for every $d, d' \in D$, the sum*

$$\sum_{y : q(y) = d'} P(x, y) \quad \text{is independent of } x \in q^{-1}(d). \tag{9}$$

In that case $\bar{P}(d, d')$ equals the displayed common value.

Proof. If $\bar{P}$ works from every concrete state, the probability of entering the abstract block d' from x must equal $\bar{P}(q(x), d')$, so the sum is constant within each fiber. Conversely, if the sums are constant, define $\bar{P}$ by that common value. Its rows sum to one, and conditional next-block probabilities depend only on the current block. Induction over observation times then yields a Markov projected process under every initial distribution. □

This is the classical strong-lumpability condition of Kemeny–Snell, not a novel MUSIC theorem [7,8]. Applying it to dreams requires an actual *state* abstraction and transition kernel; ordinary report-union episodes do not automatically satisfy either premise.

6 Composers, conductors, and computability types

A *human composer* designs the instruments, original ATP programs, proof rules, and admissible strategy space. An *algorithmic composer* is a computable mechanism that selects or transforms permitted

performance plans. An *adaptive composer* updates future plans from authenticated scores; a *stochastic composer* additionally consumes a declared random source. The *conductor* coordinates an execution of the current plan. These are proposed role distinctions, not new logical inference rules.

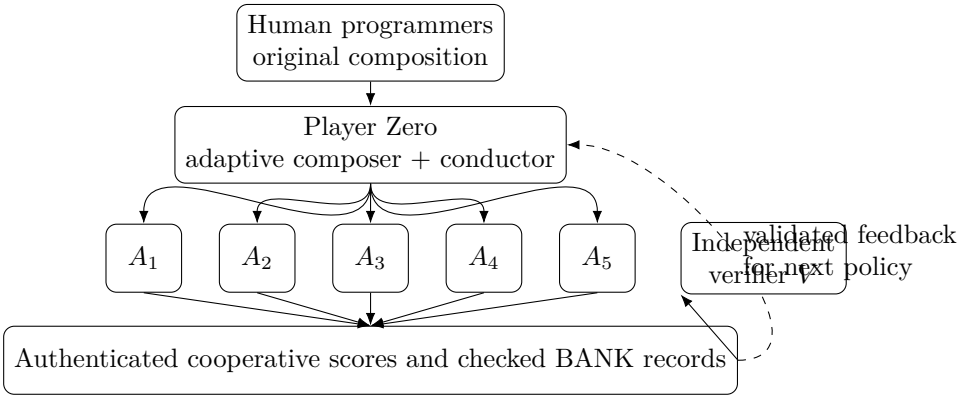


Figure 2: The proposed composer–conductor architecture. The checker is a separate authority; feedback may change search strategy but not proof-admission rules. The arrows are conceptual, not evidence of deployment.

Three computational questions must not be conflated: (a) whether a *particular finite score* can be generated or checked; (b) whether a *particular infinite stream* is computable; and (c) whether membership in the *set of attainable score labels* is decidable.

Proposition 6.1 (Effective deterministic output)**.** *Suppose a deterministic effective composer begins from computable initial data and, for every n, emits its nth symbol after finitely many effective steps. Then the resulting infinite stream is a computable sequence.*

Proof. To compute output coordinate n, simulate the algorithm until its nth output is emitted. The hypothesis ensures termination for every n. □

Theorem 6.2 (Standard algorithmic-randomness fact)**.** *A computable rule that copies independent fair bits from an ideal infinite unbiased random source outputs a Martin–Löf random infinite sequence with probability one; thus that* realized infinite output *is incomputable with probability one.*

Proof. The fair product measure of the Martin–Löf random sequences is one. No computable sequence is Martin–Löf random, since a computable sequence can be covered by a uniformly effectively enumerable test whose measure tends to zero. The copy rule remains a computable transformation of its input; the input sequence itself is almost surely noncomputable. □

This fact requires an *ideal random stream*; a computably seeded pseudorandom generator does not satisfy the conclusion merely by producing statistically convincing finite samples. It is standard background rather than a discovered ATP effect [9].

Theorem 6.3 (Undecidable attained labels with finite checkable witnesses)**.** *There exists an effective score-generation protocol whose set of labels attainable by* successful finite scores *is recursively enumerable but undecidable, even though each supplied successful score has a finite, decidably checkable computation witness.*

Proof. A legal start selects an integer program index e through a finite computable code. A participant simulates program e on empty input; a successful score carrying label e is emitted exactly when the simulation halts, with its finite halting trace attached. The remaining participants may make harmless scheduled actions. The possible attained labels are $K = \{e : \text{program } e \text{ halts on empty input}\}$, which

is recursively enumerable by dovetailing and undecidable by the halting theorem. A presented finite halting trace can nevertheless be replayed and checked step by step; this does not decide whether an arbitrary unobserved label e is ever attainable. □

The construction does not require five agents: it is a computability distinction, *not* a non-singleton indispensability result.

7 Player Zero as adaptive composer and conductor

Definition 7.1 (Frozen-contract adaptation)**.** Fix a shared object theory Γ, proof calculus F, and independent sound checker V. At stage t, let C_t be a composition policy, S_t the authenticated score of an authorized performance, and B_t the set of admitted object-level BANK formulas. Player Zero may compute

$$C_{t+1} = \Phi(C_t, S_t, B_t), \tag{10}$$

where Φ may change scheduling, strategy, role assignments, or candidate representations but *may not* revise Γ, the inference calculus, the interpretation of prior certificates, or the admission authority of V without a separately declared theory migration.

Theorem 7.2 (Verifier-gated adaptive conservativity)**.** *If $B_0 \subseteq \mathrm{Cn}_F(\Gamma)$ and each newly accepted object formula has a checker-validated derivation from $\Gamma \cup B_t$ with complete dependency scope and no unverified imports, then for every finite stage t,*

$$B_t \subseteq \mathrm{Cn}_F(\Gamma), \qquad \mathrm{Cn}_F(\Gamma \cup B_t) = \mathrm{Cn}_F(\Gamma). \tag{11}$$

This remains true regardless of how Φ changes the search policy.

Proof. Induct on accepted deposits. The base case is given. If all entries in B_t follow from Γ, a valid derivation of a new φ from $\Gamma \cup B_t$ establishes $\Gamma \vdash_F \varphi$ by substitution of the predecessor derivations, with the required scopes preserved. Therefore every new admission lies in $\mathrm{Cn}_F(\Gamma)$. Extensivity, monotonicity, and idempotence of deductive closure then yield the displayed equality. The update policy is not a premise of the verification rule. □

Remark 7.3 (Not a complete safety theorem)*.* The result says nothing by itself about search completeness, reliable policy optimization, confidentiality, real-world physical safety, nontermination, or resistance to adaptive overfitting. A self-modifying update rule Φ would require a second immutable meta-verification policy if one wanted a comparable theorem about authorized changes to the composer itself.

8 The actual Multiarticle settlement boundary

The source collection describes a small finite checker supporting quantum probability arithmetic and a deliberately narrow propositional source-dependence exercise. The current audit reproduces the probability calculation with an exact-rational script (supplementary file), but *does not* treat the earlier cited checker as available evidence: its code was not among the files attached to this request. A toy propositional model with separate source atoms and explicit bridge axioms can demonstrate how removal of an atom changes entailment, but it does not determine the deductive closures of three *complete published papers*.

To claim that a concrete physics or interface theorem T is transcendent relative to the full MUSIC, SYMPHONY, and DNI articles under Equation (3), the following must be provided:

M1. Frozen versions of each source's language, axioms, precise claim set, interpretation, and translations τ_i; the target T must have one fixed meaning across comparisons.

M2. A soundly justified bridge B that does not simply assert T, and explicit treatment of the consistency and scope of the joint theory C_M.

M3. An accepted, checkable proof of T from the translated union plus bridge, including every imported dependency.

M4. For each full individual closure C_i, a genuine non-derivability argument for T (for example a model of that translated theory falsifying T, when the calculus admits that method). An incomplete search is insufficient.

M5. If one also claims *essential contribution* of each source, additional removal tests and independently verified scope assumptions; this is stronger than membership in a set difference.

Proposition 8.1 (Shared-base closure cannot grow by rediscovery)**.** *Let* Γ *be a fixed base in a monotone, idempotent consequence system, and let* $B_i \subseteq \mathrm{Cn}_F(\Gamma)$ *contain formulas admitted only as consequences of* Γ*. Then*

$$\mathrm{Cn}_F\left(\Gamma \cup \bigcup_i B_i\right) = \mathrm{Cn}_F(\Gamma). \tag{12}$$

Proof. The left premise set contains Γ and is contained in $\mathrm{Cn}_F(\Gamma)$; applying monotonicity and idempotence yields equality. □

Cooperation can change which proofs are *found*, shared, and recorded while leaving a frozen theory's consequence set unchanged. A new theorem relative to individually restricted evidence differs from a previously unproved-to-the-searcher theorem of the common theory. The cited full-article Multiarticle transcendence claim therefore remains *unestablished*.

9 Result register and responsible publication claims

Table 2: Evidence-respecting status of the article's results. Here "proved" means argued in the stipulated model, not externally machine-certified.

Claim	Supported status	Precise scope or limitation
Five-essential Horn evidence and compact prescribed quotient	Source-derived, conditional	Evidence indispensability; centrally pooled inputs remain easy.
Local MUSIC does not determine full cooperative reachability	Source-derived proof	Pre-transfer local conditions fixed; global authorizations differ.
Affine terminal quotient from $m+1$ queries	Source-derived, conditional	Trusted affine promise; certificate-aware settlement requires additional assumptions.
Unrestricted exact black-box reconstruction from fewer than 2^m queries	Refuted universally	Unobserved input can carry an opposite label.
Five-way federation beats unrestricted centralized simulation asymptotically solely by its member count	Unsupported / contradicted under specified simulation model	Fully informed polynomial-work central simulation under effective-process assumptions.
Five-qubit $\rho_\pm$: identical proper reduced states and full support; differing joint law	Proved mathematically in Section 4	Exact-rational finite probability table reproduced in audit; not an experimental result.

Claim	Supported status	Precise scope or limitation
Unweighted MUSIC / score support determines both quantum laws	Refuted for given encoding	One support, two different measures; stochastic enrichment required.
Finite deterministic initially randomized model realizes infinite full-support word law	Refuted under Theorem 5.2	No fresh random bits, finite hidden initial states, all-horizon exact equality.
Strong Markov lumpability criterion	Standard theorem	State-based finite projection under all initial laws, not arbitrary episode abstraction.
Ideal fair-bit composer yields incomputable infinite stream almost surely	Standard theorem	Assumes true fair random input; does not apply to computable seeded PRNG.
Computable protocol can have undecidable attainable-label set	Standard reduction adapted here	Each supplied finite successful score remains individually checkable.
Adaptive composer preserves accepted-theorem closure	Conditional theorem	Fixed Γ, F, sound verifier and complete dependency checks.
Full-publication MUSIC–SYMPHONY–DNI transcendent theorem	Not established	Complete translated source closures and non-derivability certificates absent.
Authenticated five-agent deployment, empirical physics prediction, or Einstein–Peres unification	Not established	No deployed machine trace, physical experiment, or general interface theorem.

Recommended concise publication wording. *For five-qubit preparations with identical proper reduced states, we exhibit full-support local measurement outcomes whose complete joint distributions differ. This demonstrates the insufficiency of unweighted possible-dream support and motivates a probabilistic enrichment of SYMPHONY. Under an explicitly finite, initially randomized deterministic realization model we also prove an all-horizon support-counting obstruction. A fixed independent proof checker can preserve logical soundness while an adaptive composer changes search policy. Claims of full-source Multiarticle transcendence, five-ATP computational superiority, and new physics remain open.*

Language to avoid. The available record does *not* establish that five independently running ATPs have produced an externally verified theorem novel under the complete closures of MUSIC, SYMPHONY, and DNI. Nor does the quantum example by itself resolve a deterministic interpretation of physics.

10 PLAN FOR FUTURE RESEARCH: self-composing, certifiable symphonic physics

The next program is deliberately prominent because it is the forward-looking purpose of this collection, not a theorem already delivered.

10.1 Priority A: type probabilistic MUSIC and SYMPHONY

Specify a finite or measurable run space, its experimental probability law, the report map d, the score encoding S, and pushforward measures. Preserve the original powerset definition of MUSIC and give a separate name to its stochastic enrichment. Prove precisely when report or state quotients preserve

transition probabilities. Report counterexamples with equal support but different measures and state when strong lumpability applies.

10.2 Priority B: certify genuine Multiarticle transcendence

Freeze complete source versions of MUSIC, SYMPHONY, and DNI; declare translations and bridge assumptions; prepare a checked combined proof and independently reviewable models or other non-derivability arguments for every individual source closure. Test bridge-only and inconsistent-union failure modes. Prefer an actual machine-readable proof object, not a narrative of five analytical roles.

10.3 Priority C: compare adaptive composing against strong controls

Hold the theorem, checker, initial evidence, and total resource accounting constant while comparing: (i) a fixed score plan; (ii) adaptation with score feedback; (iii) adaptation without musical indexing; (iv) each four-of-five ablation; and (v) an equally informed centralized simulator. Log candidate selection, successful certificates, communication, storage, verification, and wall time. Distinguish restricted-access advantage from equal-information performance.

10.4 Priority D: specify a computability hierarchy for performances

Separate checking a given finite score, generating an infinite score, sampling from an externally randomized composer, and deciding whether an arbitrary label can ever be reached. Identify the representation and quantifiers in each decidability claim. Never infer uncomputable output from deterministic effective composition on computable data, nor infer an undecidable finite checker from undecidable *attainability*.

10.5 Priority E: a restricted Einstein–Peres interface

Begin with a small quantum observable model, declared preparations, measurement contexts, and explicit relativistic causal or signal-locality constraints. An interface should preserve both probability predictions and the stated causal requirements under declared translations. This is a candidate study of *compatibility*, not an assumption that a full physical unification exists. Extend toward curved spacetime or dynamical gravity only after a restricted result is correctly formalized and independently checked.

10.6 Flagship open questions

1. **Emergent Symphonic Quotient Discovery.** Can five interface-constrained ATPs discover and certify a nontrivial, efficiently liftable settlement quotient from cooperative dreams, with exactly stated architecture-relative necessity of each participant?
2. **Emergent Certified-Interface Discovery.** Can the federation infer a reusable cross-theory interface and independently establish a theorem meeting Equation (3) for the *full* source closures?
3. **Self-Composing SYMPHONY.** Can a verifier-gated adaptive composer improve certified discoveries per total charged cost over strong, fixed-composition baselines on a specified infinite task family?
4. **Quantum–Relativistic Transfer.** Can a concrete necessary or sufficient compatibility condition be certified between a declared quantum measurement theory and a specified relativistic observable theory?

10.7 Proposed shape of the eventual combined article

A distinct, full research paper could begin with provenance-preserving local dreams, develop legal five-ATP scores and certified quotient navigation, then prove a properly typed quantum probability example and deterministic-interface boundaries before analyzing adaptive composition. It should end with a complete-source transcendence audit and an explicit, modest physical interface theorem or a decisive counterexample. The present collection is an audited *research sourcebook*, not a substitute for those additional results.

11 Audit method, source notes, and bibliography

The attached source was an 11-page working research collection. This edition reconstructs it from that PDF, retains its mathematical storyline and limitations, repairs typographic mathematics, and provides direct proofs rather than relying solely on a reportedly executed but unavailable source checker. An exact-rational, standalone finite checker was run in the current audit: it verified the 32×32 density matrices and proper reduced states, the Born probabilities p_σ, outcome marginals, parity expectations, and TV distance; the coalition necessity of the Horn witness for $n = 1, 2, 3$; and affine identification for $m = 2, 4, 5$. These bounded tests do not certify the universal arguments or the bibliography. A detailed repair ledger accompanies the PDF as a separate audit artifact.

[**1**] B. Tenneson, *MUSIC of an Automated Theorem Prover: Possible Dream Families, Realizability Complexes, and Certified Reflection*, version 2.0, 3 November 2026. https://btenneson.github.io/pub/cs.LO_Logic_in_Computer_Science/music_of_an_atp_expository_research_article_2_0/.

[**2**] B. Tenneson, *Symphonies of Automated Theorem Provers: Cooperative Dream Histories, Certified Gestalt, and the Limits of Federated Advantage*, version 2.0, 3 November 2026. https://btenneson.github.io/pub/cs.LO_Logic_in_Computer_Science/symphonies_of_automated_theorem_provers_2_0/.

[**3**] B. Tenneson, *Deterministic–Nondeterministic Interfaces: A Textbook-Ordered Reader of Core Results, Zip–Inverse-Zip Structures, and Mathematical Audit*, 2026. https://btenneson.github.io/pub/cs.LO_Logic_in_Computer_Science/deterministic_nondeterministic_interfaces_textbook_ordered_reader_20261020/.

[**4**] B. Tenneson, *Multiarticle Theory: A Short Foundational Text*, version 0.1, 21 October 2026. https://btenneson.github.io/pub/cs.LO_Logic_in_Computer_Science/multiarticle_theory_foundations_0_1/. The present edition reproduces the comparison formula as stated in the *attached collection*; it has not authenticated the full source formalism.

[**5**] *From MUSIC to Emergent Symphonic Compression: Constructive Theorems, Discovery Limits, and a Plan for Future Research*, working research exposition, 3 November 2026; predecessor document identified in the source collection (a later audited version 2.0 was separately prepared).

[**6**] `music_symphony_dni_finite_checker.py`, companion checker named in the original collection but *not attached to this request*; its prior execution is not independently asserted here. A new finite diagnostic script `Afternoon_Research_Finite_Audit_v2.py` is included among the present deliverables.

[**7**] J. G. Kemeny and J. L. Snell, *Finite Markov Chains*, Springer reissue, 1976 (original edition 1960), strong lumpability criterion (chapter/section numbering edition-dependent).

[**8**] G. Wolfer and S. Watanabe, "Geometric aspects of data-processing of Markov chains," *Transactions of Mathematics and Its Applications* 8(1) (2024), Article tnae001, Theorem 3.4. https://doi.org/10.1093/imatrm/tnae001.

[**9**] A. Nies, *Computability and Randomness*, Oxford University Press, 2009. Standard reference for effective randomness and Martin–Löf randomness.

Full AI-Assistance and Mathematical Reliability Disclaimer

Final-page statement

This article is a human-originated, AI-assisted research exposition. Brian Tenneson supplied the overarching MUSIC, SYMPHONY, DNI, Multiarticle, AMLD+0, and composer–conductor program and the related research objectives, prior manuscript context, and intended mathematical interfaces. ChatGPT (GPT-6) assisted in rebuilding the manuscript from the attached PDF; clarifying assumptions, definitions, and the boundaries of results; checking finite probability and logic examples; drafting revised proofs, diagrams, source descriptions, and the audit record; and typesetting the resulting Version 2.0. These activities are editorial and mathematical assistance, not evidence of separately operating, authenticated Pioneer agents or an independent formal verification service.

The exposition aims to be factually accurate, logically coherent, and sufficiently complete in the ordinary editorial sense of presenting relevant facts, assumptions, reasoning, and limitations. It provides no guarantee that every claim is correct, consistent, provable, novel, or free from omitted hypotheses. The term "complete" does not assert completeness in the technical sense of mathematical logic. Proofs described as established are arguments within the formal assumptions displayed here; no proof assistant or external automated theorem prover independently certified them. Finite tests, even exact-rational exhaustive checks of restricted instances, do not replace universal proofs or outside refereeing. Bibliographic metadata and interpretations of the author's source theory should be verified against the original publications where they matter.

Nothing here establishes that a physically deployed five-ATP AMLD+0 federation ran the protocols, that MUSIC or SYMPHONY generates greater theoremhood, that an empirical proof-search speedup has been measured, or that a new law of physics has been discovered. The quantum example is a constructed finite-dimensional state calculation, not a new experiment. The deterministic-interface obstruction is restricted to finite hidden initial-state models with no subsequent randomness; it is not a theorem of fundamental physical indeterminism. The Multiarticle closure comparison is source- and translation-relative, and no full-paper transcendence certificate or Einstein–Peres unification is claimed.

Readers should inspect the mathematics and cited sources, validate any intended implementation's checker and dependency scopes, and seek independent verification before relying on the results in scientific, legal, engineering, or other consequential settings. The present iterative "Atlas audit" was performed as an AI-assisted manuscript inspection and repair, not as an external mathematical peer review.

End of manuscript